17/4/2026

Recall

$$\int_0^{\pi/2} \sin^p\theta \cos^q\theta \, d\theta = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

Eg: Evaluate $\displaystyle\int_0^\infty \frac{x^2}{(1+x^6)^{7/2}}\,dx$

Soln: put $x^3 = \tan\theta \Rightarrow x^6 = \tan^2\theta$

$\Rightarrow 3x^2\,dx = \sec^2\theta\,d\theta$

x	0 to ∞
$\theta = \tan^{-1}(x^3)$	0 to $\pi/2$

$$\therefore \int_0^\infty \frac{x^2\,dx}{(1+x^6)^{7/2}}$$

$$= \int_0^{\pi/2} \frac{\frac{1}{3}\sec^2\theta\,d\theta}{(1+\tan^2\theta)^{7/2}}$$

$$= \frac{1}{3}\int_0^{\pi/2} \frac{\sec^2\theta}{(\sec^2\theta)^{7/2}}\,d\theta$$

$$= \frac{1}{3}\int_0^{\pi/2} \frac{\sec^2\theta}{\sec^7\theta}\,d\theta$$

$$\Rightarrow I = \frac{1}{3}\int_0^{\pi} \frac{1}{\sec^5\theta}\,d\theta$$

$$= \frac{1}{3}\int_0^{\pi/2} \cos^5\theta\,d\theta$$

$0 = \sin^0\theta \cos^5\theta$

$$= \frac{1}{3}\cdot\frac{1}{2}\beta\left(\frac{1}{2}, 3\right)$$

$$= \frac{1}{6}\,\frac{\Gamma(\frac{1}{2})\cdot\Gamma(3)}{\Gamma(7/2)}$$

$$= 8/45$$

Revision.

1. $$\Gamma(k) = \begin{cases} \int_0^\infty e^{-x} x^{k-1}\,dx \\ 2\int_0^\infty e^{-x} x^{2k-1}\,dx \end{cases}$$

2. $\Gamma(1) = 1$

3. $\Gamma(k) = (k-1)\Gamma(k-1)$

4. $\Gamma(n) = (n-1)!$
   $n \to$ +ve integer

5. $\Gamma(p)\Gamma(1-p) = \dfrac{\pi}{\sin \pi p}$

6. $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx = \beta(n,m)$

7. $\beta(m,n) = \dfrac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

8. $\beta(m,n) = \int_0^\infty \dfrac{x^{m-1}}{(1+x)^{m+n}}\,dx$

9. $\int_0^{\pi/2} \sin^p\theta \cos^q\theta\,d\theta = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$

Set A
1404 A

Set B
1404 B

Look at the limits | Change to | Substitution

(1) $\int_0^{\pi} \sin^2\theta\,(1+\cos\theta)^3\,d\theta \rightarrow$ $\underline{0 \text{ to } \pi}$ | $\underline{0 \text{ to } \pi/2}$ | $\theta \rightarrow \theta/2$ put $\theta/2 = t$

(2) $\int_{-\pi/4}^{\pi/4} (\underline{\sin\theta} + \cos\theta)^{1/3}\,d\theta \rightarrow -\frac{\pi}{4}$ to $\pi/4$ | 0 to $\pi/2$ | $\theta + \pi/4$

$\sin(\theta + \pi/4)$

$(\sin\theta \cdot \cos\pi/4 + \cos\theta \sin\pi/4) \cdot \frac{1}{1/\sqrt{2}}$: $\cos\pi/4 = \sin\pi/4 = \frac{1}{\sqrt{2}}$

$\frac{1}{\sqrt{2}} = \sin\pi/4 = \cos\pi/4$ $\quad t = \theta + \pi/4$

introduce $\pi/4$

θ : 0 to ∞

$\tan\theta$: 0 to $\pi/2$

(3) $\int_0^{\infty}$ $\quad x^3 = \tan\theta$ $\quad 0$ to ∞ $\quad 0$ to $\pi/2$

$\alpha\, x^2\,dx = \sec^2\theta\,d\theta$

Duplication formula for gamma fns

$$\Gamma(m)\,\Gamma\left(m+\frac{1}{2}\right) = \frac{1}{2^{2m-1}}\,\Gamma(2m)\sqrt{\pi}$$

$\Gamma\left(\frac{1}{2}\right)$

$m = \frac{1}{4}$

$$\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right) = \frac{1}{2^{1/2}-1}\,\Gamma\left(\frac{1}{2}\right)\sqrt{\pi}$$

ie $\dfrac{\pi}{\sin \pi/4} = \dfrac{\pi}{\sqrt{2}-1}$